

Lecture 20

— Midterm 2 in one week.

So far: bijections, sum, product, division, basic counts, $\binom{n+r-1}{r-1}$
binomial thm, combinatorial identities. "Stars and bars"?

I Recurrence Relations

II Pigeonhole Principle

I. ex1: Count # of strings with letters L, A of length n
with no consecutive LL.

LALALAAAAA, LA~~LL~~A

$$S_n = \{ (x_1, \dots, x_n) : x_i \in \{L, A\}, \forall i \nexists (x_i = x_{i+1} = L) \}$$

Define a sequence $a_1, a_2, a_3, \dots, a_n, \dots$ with $a_n = |S_n|$.

A recurrence relation is a recursive definition of a sequence.

Base Case :

$$a_1 = 2$$

L
A

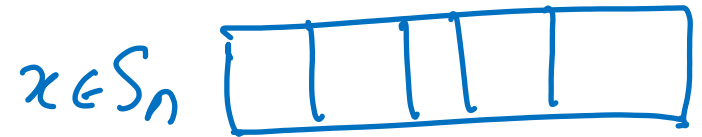
(small examples)

$$a_2 = 3$$

AA
AL
LA

Observe that for $n \geq 3$, each $x \in S_n$

either starts with A or with LA,
and these cases are mutually exclusive.



Thus,

$$S_n = S_n^A \cup S_n^{LA} \text{ where}$$

$$S_n^A = \{ (A, x_2, \dots, x_n) : (x_2 \dots x_n) \text{ has no LL} \}$$

$$S_n^{LA} = \{ (L, A, x_3, \dots, x_n) : (x_3 \dots x_n) \text{ has no LL} \}$$



$$|S_n^A| = |S_{n-1}|$$

$$|S_n^{LA}| = |S_{n-2}|$$

and these sets are disjoint.

Thus, by the sum rule

$$a_n = |S_n| = |S_n^1| + |S_n^{10}| \\ = a_{n-1} + a_{n-2}.$$

Recursive Step.

Soln:

$$\begin{array}{l} a_1 = 2 \\ a_2 = 3 \end{array} \quad a_n = a_{n-1} + a_{n-2}.$$

* Can efficiently calculate a_n : 2, 3, 5, 8, 13, 21, 34, 55, ...
 a_1 a_2 a_3 a_8

* Can prove things about a_n

(Fibonacci starting at 2, 3)

Implicit vs explicit solution (formula).

"Solving" a recurrence: getting a formula.

$$\begin{cases} f_0 = 1 & f_1 = 1 \\ f_n = f_{n-1} + f_{n-2} \end{cases}$$

Math 54
Diagonalization $\rightarrow$

$$f_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$$

Optional: Read 8.2, Math 54/185.

ex2: Let $S_n = \{ (x_1, \dots, x_n) : x_i \in \{0,1\}, \exists i (x_i = x_{i+1} = 0) \}$.

bit strings of length n containing **00**. (at least once).
Let $b_n = |S_n|$, $n \geq 1$.

examples: $\begin{cases} b_1 = 0 \\ b_2 = 1 \\ b_3 = 3 \end{cases}$

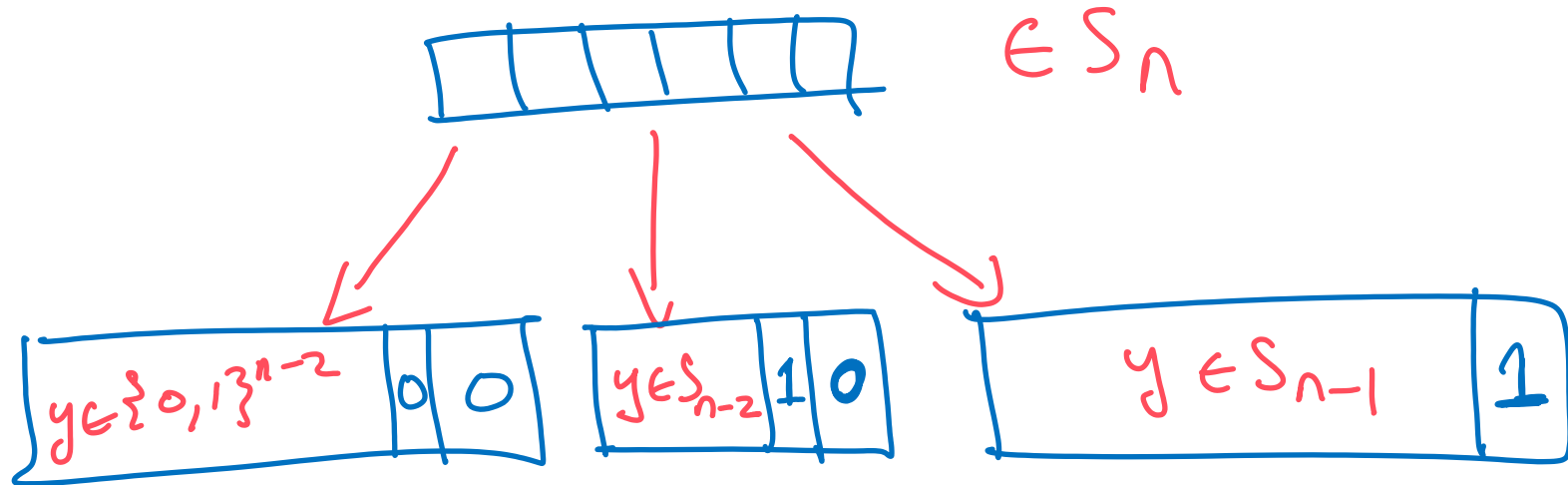
Basis
step.

00

100, 001, 000

b_4 : tack on 1 or 0
1001, 0011, 0001
1000, 0010, 0000
(1100)

Assume $n \geq 3$.



Every element $x \in S_n$ is of one of the following types:

- ① $x = (y, 0, 0)$ with $y \in \{0,1\}^{n-2}$. 2^{n-2}
- ② $x = (y, 1, 0)$ $y \in S_{n-2}$. b_{n-2}
- ③ $x = (y, 1)$ $y \in S_{n-1}$ b_{n-1}

and these cases are mutually exclusive.

By the sum rule,
 $b_n = 2^{n-2} + b_{n-2} + b_{n-1}$ for $n \geq 2$.

Recursive Defn.

Poll:

If T is a tree on n vertices, then
the # of 3-colorings of T is
not divisible by 5.

Prop:

If T is a tree on $\{1, \dots, n\}$:

the number of 3-colorings of T is independent of T . $\leftarrow$

$P(k)$ { (A)
(B)

Satisfies the recurrence

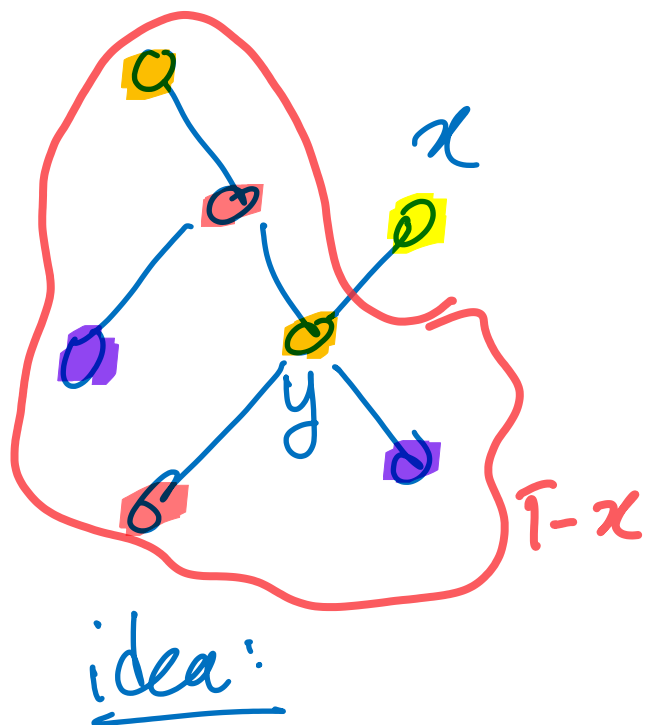
$$\begin{aligned} a_1 &= 3 \\ a_n &= 2 \cdot a_{n-1} \end{aligned} \text{ for } n \geq 2.$$

Proof: By induction.

Base case: T has one vertex, so $a_1 = 3$.

Inductive Step: Assume $n \geq 1$. Assume $P(k)$.

Assume T is a tree on $n+1$ vertices.



Choose a leaf $x \in T$. Let y be the unique neighbor of x in T .

Observe that a coloring of T is uniquely specified by the following process:

- used independence $\rightarrow$
- ① Choose a 3 -coloring of $T-x$. a_n
 - ② Choose a color $f(x)$ different from $f(y)$. 2

By the product rule,

$$a_{n+1} = a_n \cdot 2.$$

establishing
 (A)/(B) $\square$

∴

$$a_1 = 3$$

$$a_n = 2 \cdot a_{n-1}$$

$$n \geq 2$$

recurrence.

"Solving" means finding a formula:

$$a_n = 2 \cdot a_{n-1} = 2 \cdot 2 \cdot a_{n-2} \cdots = \underline{\underline{2^{n-1} \cdot 3}}.$$

5 for a_n by FTA.

II Pigeonhole Principle

Theorem: If $k \geq 1$ and $k+1$ or more pigeons ^{objects} are placed in k boxes, then there must be a box with at least 2 pigeons.

Proof: By contradiction. Assume each box has 0 or 1 pigeons. There are k boxes, so there must be at most k pigeons. $\square$

Equivalently: If $|S| \geq |T|$ and $f: S \rightarrow T$ ^{placement}
then f is not 1-1.

Prop 1: In every group of 367 people, there must be two with the same birthday.

Proof: Create 366 boxes, one for each day of the year. Place 367 people in boxes. By PP., some box has at least 2. $\square$

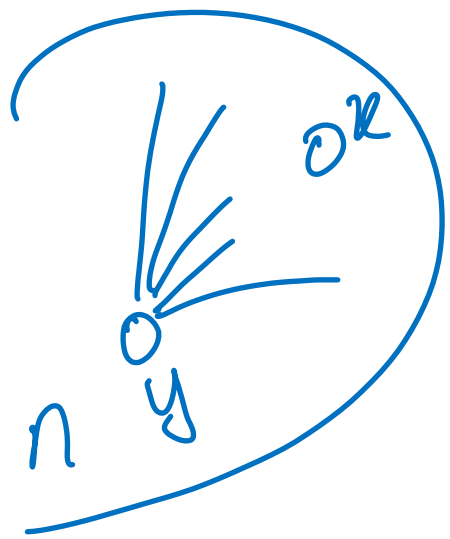
Prop 2: If 5 distinct integers are chosen from $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$, then two of them must add up to 9.

Proof: Observe that A can be partitioned into four pairs $\{1, 8\}, \{2, 7\}, \{3, 6\}, \{4, 5\}$ such that the sum in each pair is 9.

Create a box for each pair. any
choice of 5 integers in A corresponds
to a placement into these 4 boxes.
By the p.p., some box must have 2
integers, which must add up to 9 by
construction. $\square$

Prop 3: If G is a simple graph with ≥ 2
vertices, it must have two vertices
of the same degree.

Proof: Assume G is simple with $n \geq 2$ vertices.



Observe that G cannot have
 both a vertex x with $\deg(x)=0$
 and y with $\deg(y)=n-1$

— in this case, simplicity implies
 $\{\cancel{0}, 1, \dots, \cancel{n-1}\}$ that y is adjacent to every vertex
 other than itself, including x , which is absurd.

Thus, the number of distinct degrees of
 vertices in G is at most $n-1$.

Make a box for each possible degree.

Since there are n vertices, p.p. implies
 that there is a box w/ two vertices. $\square$

Remark: "nonconstructive" proof.

Prop 4 (Special case of Chinese Remainder).

Suppose $\gcd(m, n) = 1$, $0 \leq a < m$
 $0 \leq b < n$. Then,

there is an integer x s.t.

$$x \equiv a \pmod{m} \quad \text{--- (1)}$$

$$x \equiv b \pmod{n} \quad \text{--- (2)}$$

Proof:

Consider the integers

$$a_0 = a, a_1 = a + m, a_2 = a + 2m, \dots, a_{n-1} = a + (n-1)m.$$

Notice that $a_i \equiv a \pmod{m}$ for all $i = 0, \dots, n-1$.

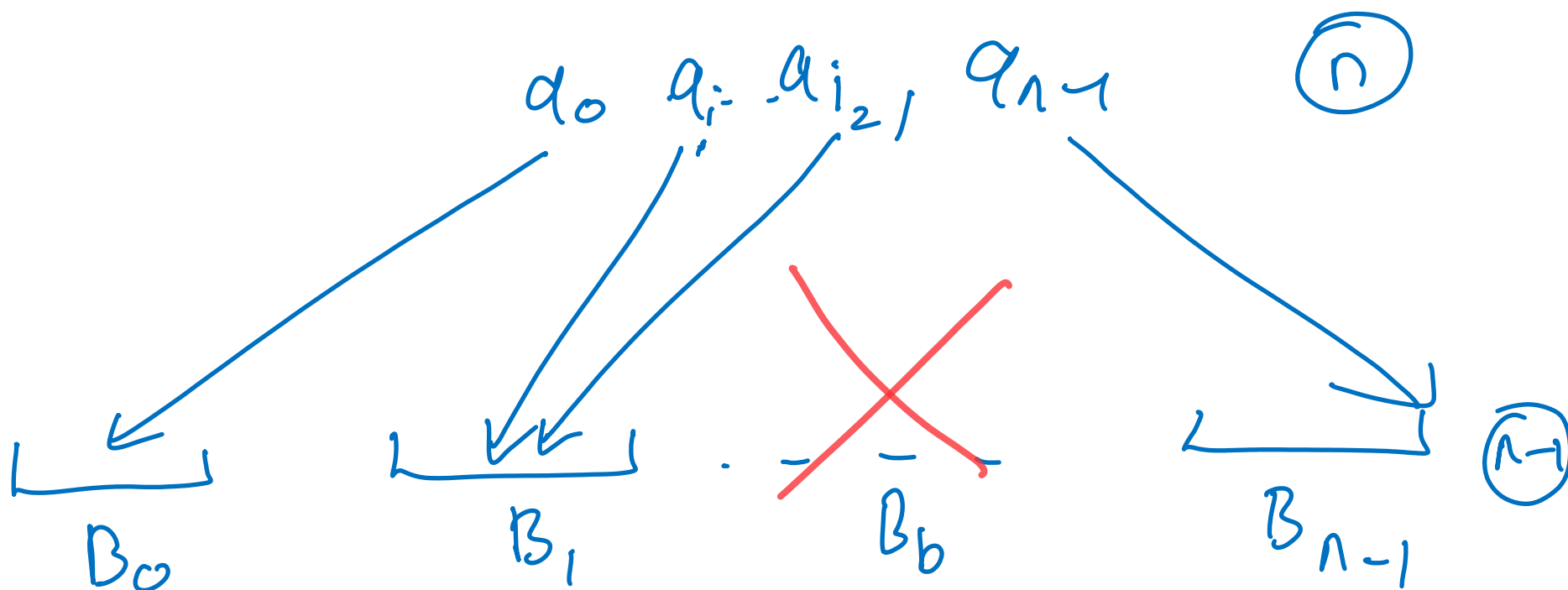
$a_0 \dots a_{n-1}$

Assume for contradiction $a_i \not\equiv b \pmod{n}$
for all $i=0 \dots n-1$.

Define n boxes

$$B_j = \{x \in \mathbb{Z} : x \equiv j \pmod{n}\}.$$

Notice that each $a_i \in B_j$ for some j .



By the p.p. there is a box B_j and $i_1 \neq i_2$ such that

$$a_{i_1} \in B_j \text{ and } a_{i_2} \in B_j.$$

This means $a_{i_1} \equiv a_{i_2} \equiv j \pmod{n}.$

$$\implies n \mid a_{i_1} - a_{i_2}$$

$$\implies n \mid a + i_1 m - (a + i_2 m)$$

$$\implies n \mid (i_1 - i_2)m$$

(Euclid)
 $\implies$

$$n \mid i_1 - i_2$$

which is absurd
since $i_1, i_2 < n$. $\square$